

# MGMT 682: Digital Product Design

## Assignment 3:A/B Testing, Continued Development, and Designing Algorithms

Please write your answers as clearly and concisely as possible. Please type your answers. Recall the course policy on AI: you are free to use AI tools subject to disclosure when and how it was used and you are fully responsible for everything you turn in.

You work an online content platform (e.g. YouTube) on the recommendations team. Your team has been tasked with re-designing recommendations that will maximize the user's long-term value/happiness from scratch. You have access to an amazing team of engineers who can train complicated models to predict different variables. That is, let's assume we can perfectly learn conditional expectation functions  $E[Y|X = x] = f(x)$ . Remember, these are fancy ways to form the best prediction we can about some outcome  $Y$  given data  $x$ .

1. Designing recommendations. (20 points)
  - a. Recall many recommendation systems predict engagement rates and recommend content in descending order of predicted engagement. Explain (1) why this is a reasonable starting point and (2) whether you think this will be sufficiently aligned with long-term user value. (10 points)
  - b. Your content moderation team worries about users falling down rabbit holes of low-quality content and you have been tasked with improving the recommendation system to prevent this. How might you modify the recommendation algorithm to combat this concern? How do you think this would impact engagement? (5 points)
  - c. How else might you modify your recommendation algorithm to better align with long-term user value? (5 points)
2. We will now test our new algorithm against the existing algorithm using A/B tests. (20 points)
  - a. Describe how you would randomize units into treatment and control. What is the unit of randomization? What share of users would you treat? (5 points)
  - b. How would you test that the randomization worked as expected? Be as specific as you can about what null hypothesis you would test. (5 points)
  - c. How would you evaluate the performance of the new algorithm against the existing algorithm? (5 points)
  - d. Describe a procedure for estimating treatment effect heterogeneity. What subgroups would you focus on? What concerns could arise when you estimate many heterogeneous treatment effects? (5 points)

## ANSWER

### 1. Designing Recommendations

#### (a) Initial Recommendation Strategy: Predicting Engagement

##### i. Explain why this is a reasonable starting point

Predicting engagement rates (e.g., clicks, time spent) and recommending content based on this prediction is a reasonable initial strategy because **engagement signals are easy to collect** and are highly sensitive indicators. Furthermore, engagement is directly related to customer value and maximizing it aligns with maximizing exposure for advertising revenue.

##### ii. Whether this will be sufficiently aligned with long-term user value

Relying solely on predicted engagement will **not** be sufficiently aligned with maximizing the user's long-term value or happiness. Engagement functions as a short-term driver metric, which is often an imperfect measure of the true long-term organizational objective. This single focus risks promoting content that users find "sensationalist and provocative," which, even if engaging in the short term, threatens long-term satisfaction and retention.

#### (b) Modifying the Algorithm to Combat Low-Quality Rabbit Holes

##### i. How might you modify the recommendation algorithm to combat this concern

To prevent users from consuming low-quality content, the algorithm should be modified into a **quality-aware ranking system**. Utilizing the team's predictive capability ( $E[Y|X=x]=f(x)$ ), the algorithm must explicitly incorporate a prediction of content quality ( $Q$ ) alongside predicted engagement ( $E$ ). The rank score ( $S$ ) should be a weighted average:  $S = wE * E + wQ * Q$  (Where  $wE$  and  $wQ$  are the assigned weights for Engagement and Quality, respectively.)

##### ii. How would this impact engagement

Implementing a strong positive weight toward quality would likely result in a **reduction in average immediate engagement** compared to a system optimized only for clicks. This trade-off is accepted to improve the long-term health of the user experience.

#### (c) How else might you modify your recommendation algorithm to better align with long-term user value?

To maximize alignment with long-term user happiness, the focus should shift to maximizing an **Overall Evaluation Criterion (OEC)**:

i. **Target Predicted Long-Term Outcomes:** The algorithm should use the predictive function  $E[Y|X=x]=f(x)$  to maximize the **predicted conditional expectation of a long-term metric (Y)**, such as future retention or lifetime value, estimated from short-term data (X).

ii. **Incorporate Explicit Satisfaction Signals:** Since behavioral data like clicks are imperfect proxies for happiness, explicit feedback should be gathered (e.g., surveying a subset of users: "Is this post worth your time?") and then modeled using prediction capabilities to scale satisfaction across the entire platform.

- iii. **Establish Penalizing Guardrails within the OEC:** The OEC should be refined to include penalties against long-term negative events. For example, the OEC could penalize the predicted probability of a user churning or unsubscribing, ensuring that short-term gains are offset if they risk long-term value loss.

## 2. A/B Tests

**(a) Describe how you would randomize units into treatment and control. What is the unit of randomization? What share of users would you treat?**

- i. **Randomization Procedure and Unit:** We must use genuine **random assignment** to split users between the control (existing algorithm) and the treatment (new algorithm). The primary unit of randomization should be the **user**, as this allows for accurate measurement of the causal effect on individual long-term value.
- ii. **Share of Users Treated:** A **50/50 split** between the control and treatment groups is typically recommended to maximize statistical power.

**(b) How would you test that the randomization worked as expected? Be as specific as you can about what null hypothesis you would test.**

- i. **Testing Procedure:** We conduct **balance tests** to compare the average values of characteristics measured **before** the experiment starts. This confirms the groups were statistically similar before the intervention.
- ii. **Specific Null Hypothesis:** For any pre-treatment characteristic (Y), we test whether the treatment indicator (D) affects that characteristic. The model tested is  $Y = a + b * D + e$ . The specific null hypothesis (H0) is that the coefficient on the treatment indicator (b) is zero: **H0: b = 0**.

**(c) How would you evaluate the performance of the new algorithm against the existing algorithm?**

Evaluation focuses on estimating the **causal impact** (treatment effect, T) on the primary success metric. This metric must be the defined **Overall Evaluation Criterion (OEC)**, which encodes the organizational trade-offs (e.g., quality vs. engagement). The treatment effect is calculated as the difference in the average OEC outcome (Y) between the two groups:

$$T = E[Y | D=1] - E[Y | D=0] \text{ (Where } E \text{ represents the average outcome in that group.)}$$

**(d) Describe a procedure for estimating treatment effect heterogeneity. What subgroups would you focus on? What concerns could arise when you estimate many heterogeneous treatment effects?**

- i. **Procedure for Estimating Heterogeneity:** Heterogeneity is estimated by calculating the treatment effect among specific user **subgroups** separately, or by including **interaction terms** between the treatment variable and subgroup characteristics in the statistical analysis.
- ii. **Subgroups to Focus On:** Key subgroups should be those related to long-term value and content problems, such as users grouped by **engagement level** (e.g., light users versus moderate users) and users identified as highly susceptible to consuming **low-quality or sensationalist content**.
- iii. **Concerns with Estimating Many Heterogeneous Treatment Effects:** The major concern is **Multiple Hypothesis Testing**. Testing numerous subgroups increases the probability of generating a **false positive** result purely by chance.